



14 COORDINATION & RESPONSE

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[VIEW EXAM QUESTIONS](#)

YOUR NOTES



14.1 NERVOUS CONTROL IN HUMANS

The Nervous System

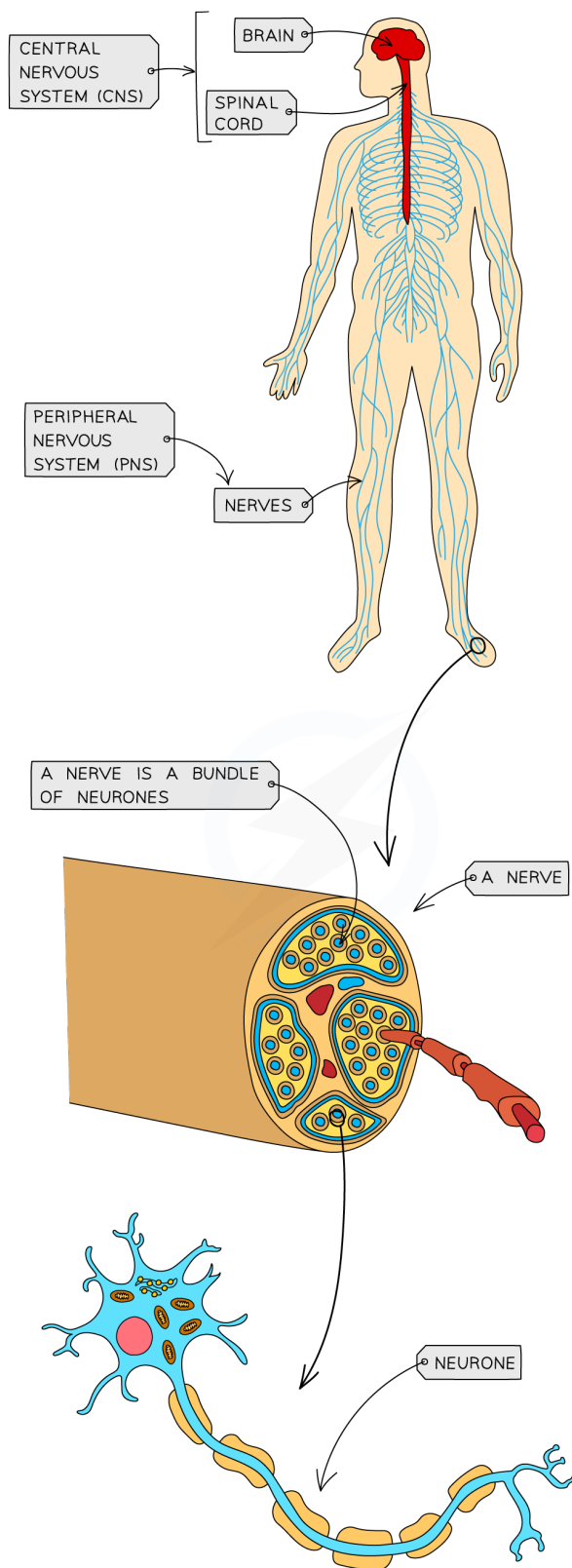
- The **human nervous system** consists of the:
 - **central nervous system** (CNS) – the brain and the spinal cord
 - **peripheral nervous system** (PNS) – all of the nerves in the body
- It allows us to make sense of our surroundings and respond to them and to **coordinate and regulate body functions**
- Information is sent through the nervous system as **nerve impulses** – electrical signals that pass along nerve cells known as **neurones**
- A bundle of neurones is known as a **nerve**



14 COORDINATION & RESPONSE

14.1 NERVOUS CONTROL IN HUMANS cont...

YOUR NOTES



The human nervous system



14 COORDINATION & RESPONSE

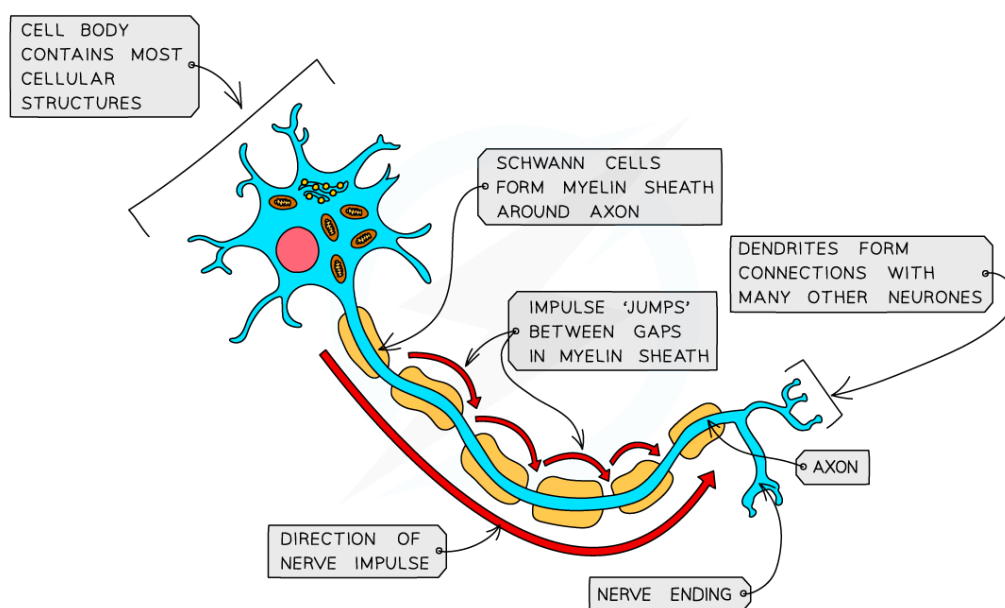
14.1 NERVOUS CONTROL IN HUMANS cont...

YOUR NOTES



Types of Neurone

- There are three main types of neurone: **sensory, relay and motor**
- Sensory neurones carry impulses **from sense organs to the CNS** (brain or spinal cord)
- **Relay** neurones are found inside the CNS and **connect sensory and motor neurones**
- **Motor** neurones carry impulses **from the CNS to effectors** (muscles or glands)
- Neurones have a long fibre (**axon**)
- This means that less time is wasted transferring the impulse from one cell to another
- The axon is insulated by a fatty sheath with small uninsulated sections along it (called nodes)
- This means that the electrical impulse does not travel down the whole axon, but **jumps from one node to the next**
- Their cell body contains many extensions called **dendrites**
- This means they can **connect to many other neurones** and receive impulses from them, forming a network for easy communication



A neurone



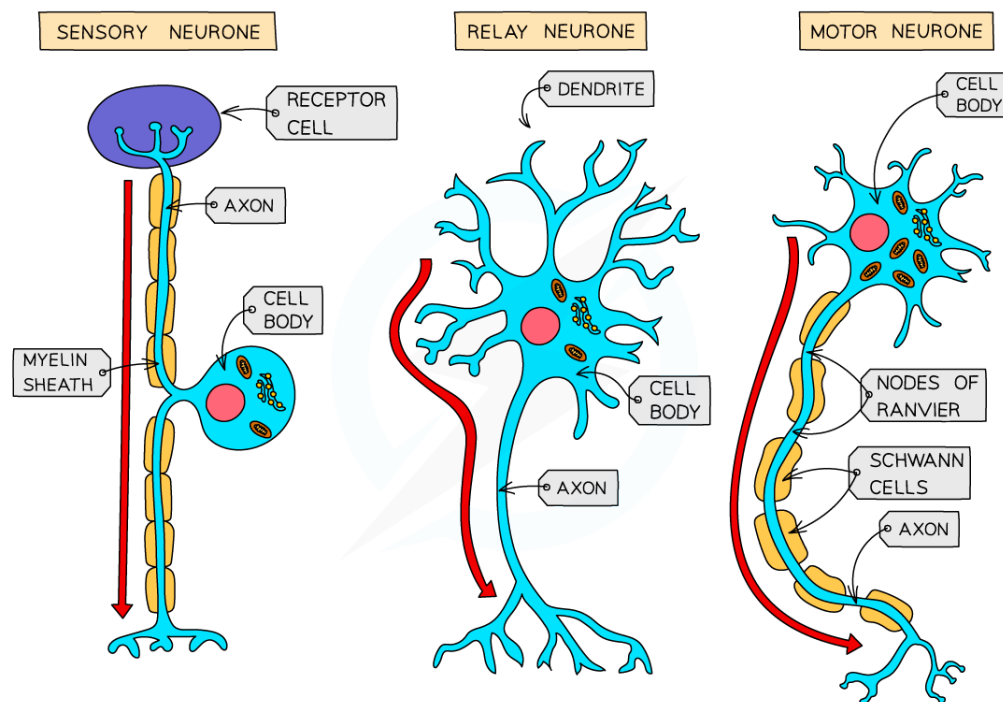
14 COORDINATION & RESPONSE

14.1 NERVOUS CONTROL IN HUMANS cont...

YOUR NOTES



Identifying the types of neurone:



The three types of neurone

- **Sensory neurones** are **long** and have a **cell body branching off the middle of the axon**
- **Relay neurones** are **short** and have a **small cell body at one end** with many dendrites branching off it
- **Motor neurones** are **long** and have a **large cell body at one end with long dendrites branching off it**



EXTENDED ONLY

Voluntary & Involuntary Responses

- A voluntary response is one where you make a **conscious decision** to carry out a particular action therefore it **starts with your brain**
- An involuntary (or reflex) response does **not involve the brain as the coordinator** of the reaction and you are **not aware** you have completed it until **after** you have carried it out
- Involuntary actions are usually ones which are **essential to basic survival** and are **rapid**, whereas voluntary responses often **take longer** as we consider what the response might be before doing it



14 COORDINATION & RESPONSE

14.2 REFLEXES

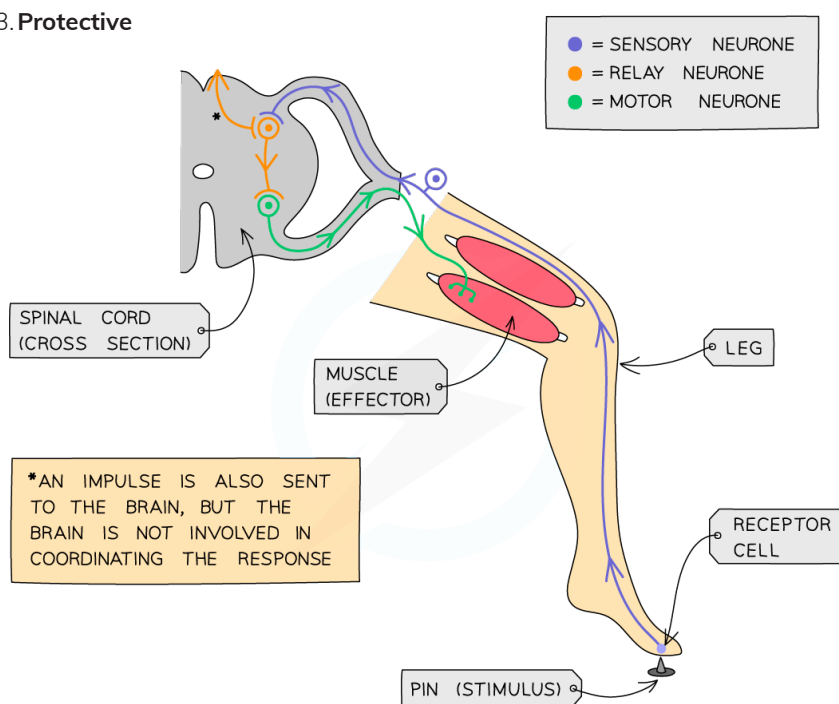
YOUR NOTES



The Reflex Arc

- An **involuntary** (or **reflex**) response **does not involve the brain** as the coordinator of the reaction and you are not aware you have completed it until after you have carried it out
- This is an **automatic and rapid response** to a stimulus such as touching something sharp or hot
- As it does not involve the brain, a reflex response is **quicker** than any other type of nervous response
- This helps to **minimise the damage to the body**
- **Reflex actions are:**

1. **Automatic**
2. **Fast**
3. **Protective**



A reflex

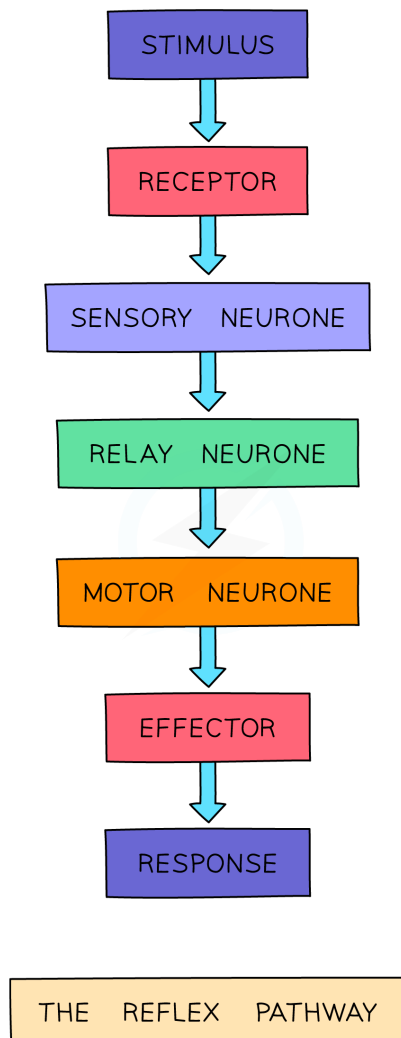
4. The pin (the stimulus) is detected by a pain/pressure/touch receptor in the skin
5. Sensory neurone sends electrical impulses to the spinal cord (the coordinator)
6. Electrical impulse is passed on to relay neurone in the spinal cord
7. Relay neurone connects to motor neurone and passes the impulse on
8. Motor neurone carries impulse to a muscle in the leg (the effector)
9. The muscle will contract and pull the foot up and away from the sharp object (the response)



14 COORDINATION & RESPONSE

14.2 REFLEXES cont...

YOUR NOTES



The reflex pathway



EXAM TIP

A common exam question is to be asked to draw arrows on the neurones in the reflex arc diagram to show the direction of movement of the impulse.

Make sure you read questions carefully – not all questions have a line underneath them to write an answer!



14 COORDINATION & RESPONSE

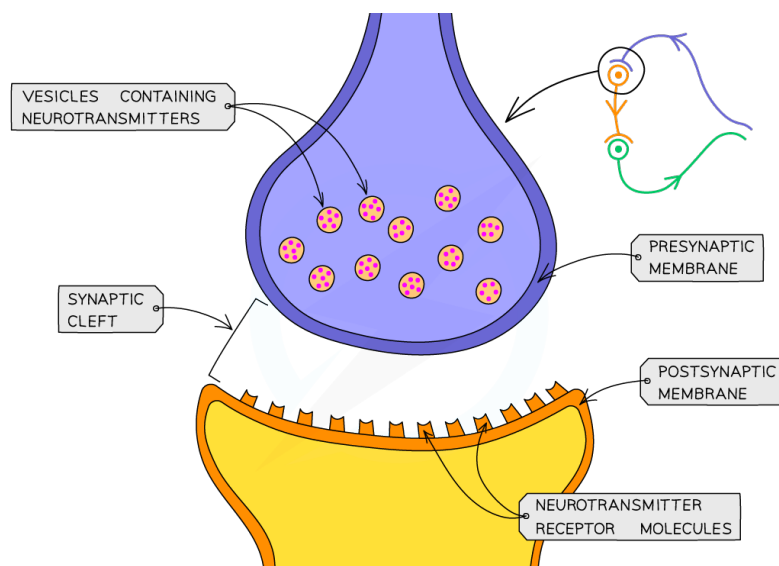
14.2 REFLEXES cont...

YOUR NOTES



The Synapse: Definition

The junction between two neurones is known as a **synapse**



A synapse



EXTENDED ONLY

How an Impulse is Passed Across a Synapse

- Neurones never touch each other
- The junctions (gaps) in between them are called **synapses**
- The electrical impulse travels along the first axon
- This triggers the nerve-ending of the presynaptic neurone to release **chemical** messengers called **neurotransmitters** from vesicles which fuse with the presynaptic membrane
- The neurotransmitters **diffuse** across the synaptic gap and **bind with receptor molecules** on the membrane of the second neurone (known as the **post synaptic membrane**)
- This **stimulates** the second neurone to generate an electrical impulse that travels down the second axon
- The neurotransmitters are then **destroyed to prevent continued stimulation** of the second neurone which would cause repeated impulses to be sent
- Synapses ensure that impulses **only travel in one direction**, avoiding confusion within the nervous system if impulses were travelling in both directions
- As this is the only part of the nervous system where messages are **chemical** as opposed to electrical, it is the **only place where drugs can act to affect the nervous system** – eg this is where heroin works



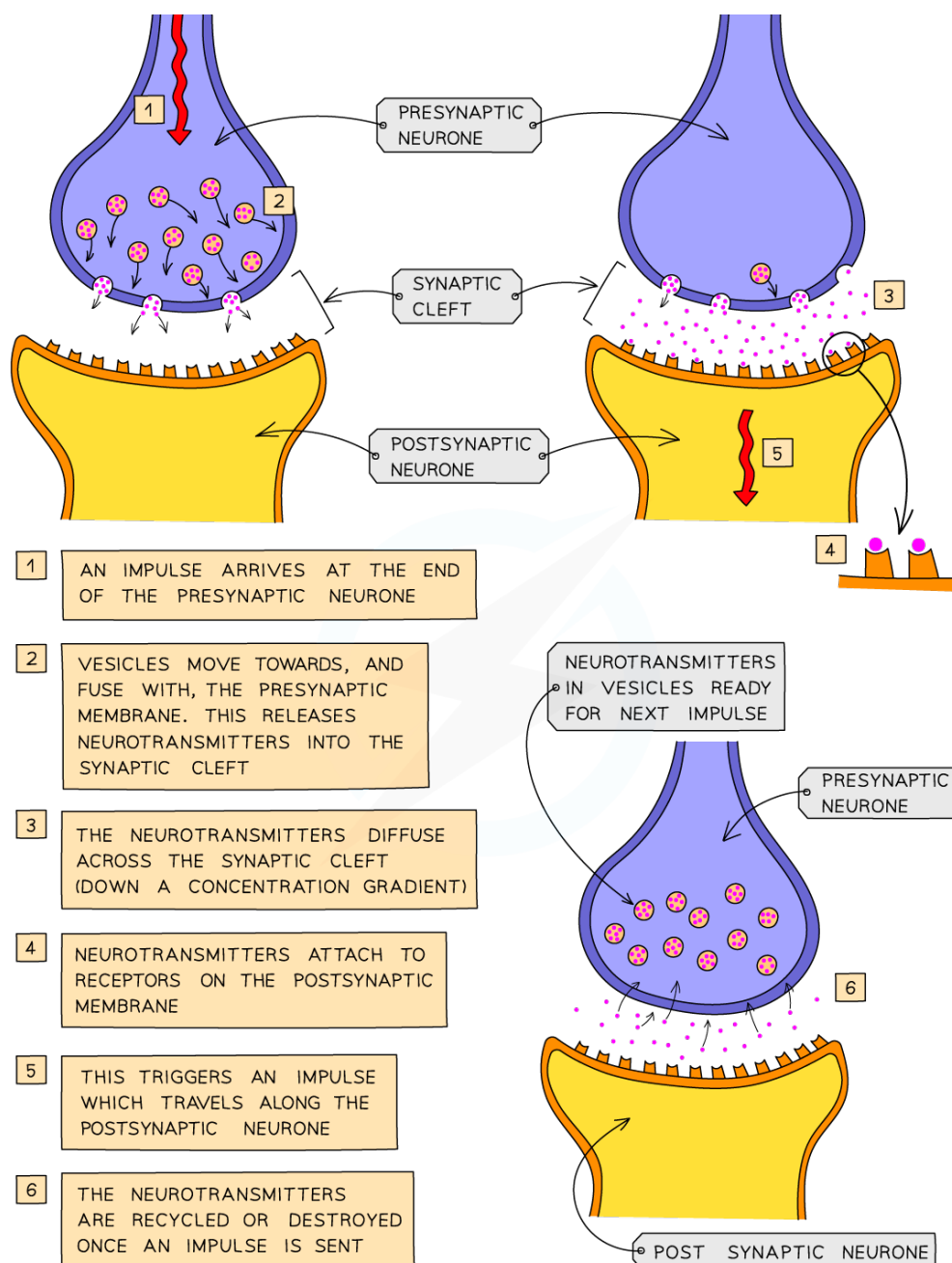
14 COORDINATION & RESPONSE

14.2 REFLEXES cont...

YOUR NOTES



EXTENDED ONLY cont...



How an impulse is passed on at a synapse



14 COORDINATION & RESPONSE

14.2 REFLEXES cont...



EXAM TIP

For maximum marks you will need to be able to understand the **structure** and **functioning** of a synapse and explain **what happens at each step**.

YOUR NOTES



14.3 SENSE ORGANS: THE EYE

Sense Organs

- Receptors are groups of specialised cells
- They **detect a change in the environment and stimulate electrical impulses in response**
- Sense organs contain **groups of receptors** that respond to specific stimuli

SENSE ORGAN	WHAT IT IS SENSITIVE TO	SENSE
SKIN	SENSITIVE TO PRESSURE, HEAT AND COLD (TEMPERATURE) AND PAIN.	TOUCH AND TEMPERATURE
TONGUE	SENSITIVE TO CHEMICALS IN FOOD AND DRINK	TASTE
NOSE	SENSITIVE TO CHEMICALS IN THE AIR	SMELL
EAR	SENSITIVE TO SOUND AND MOVEMENT	a) HEARING b) BALANCE
EYE	SENSITIVE TO LIGHT	SIGHT

- Once the receptor cell in the sense organ has been stimulated, it generates an electrical impulse
- This is passed on to a sensory neurone which carries the impulse to the central nervous system
- Here a response will be decided on and the impulse will be passed to a motor neurone (via a relay neurone)
- The motor neurone carries the impulse to the effector (muscle or gland)
- The effector carries out the response



14 COORDINATION & RESPONSE

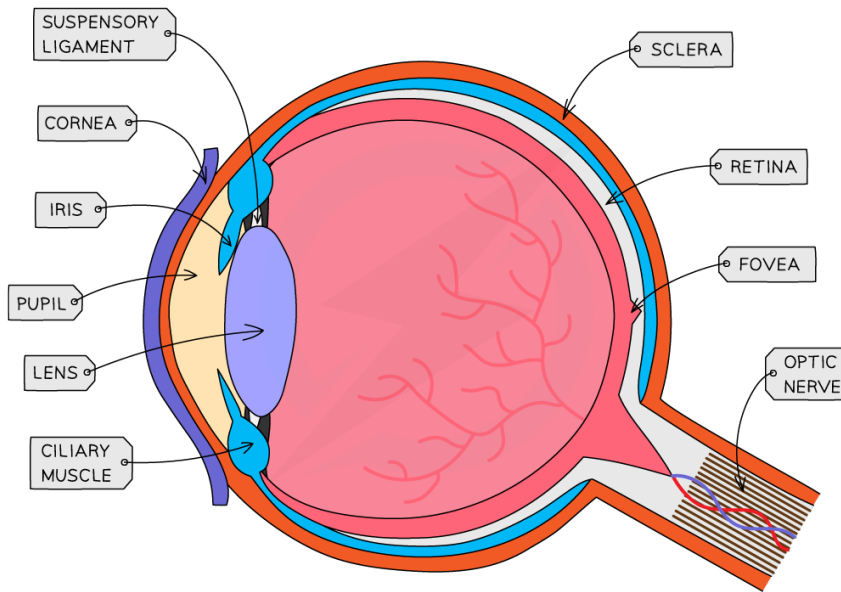
14.3 SENSE ORGANS: THE EYE cont...

YOUR NOTES



Structure of the Eye & Functions

- The eye is a **sense organ** containing receptor cells that are sensitive to **light** (rod cells) and **colour** (cone cells)



The eye

Function of the parts of the eye

STRUCTURE	FUNCTION
CORNEA	TRANSPARENT LENS THAT REFRACTS (BENDS) LIGHT AS IT ENTERS THE EYE
IRIS	CONTROLS HOW MUCH LIGHT ENTERS THE PUPIL
LENS	TRANSPARENT DISC THAT CAN CHANGE SHAPE TO FOCUS LIGHT ONTO THE RETINA
RETINA	CONTAINS LIGHT RECEPTOR CELLS – RODS (DETECT LIGHT INTENSITY) AND CONES (DETECT COLOUR)
OPTIC NERVE	SENSORY NEURONE THAT CARRIES IMPULSES BETWEEN THE EYE AND THE BRAIN
PUPIL	HOLE THAT ALLOWS LIGHT TO ENTER THE EYE



14 COORDINATION & RESPONSE

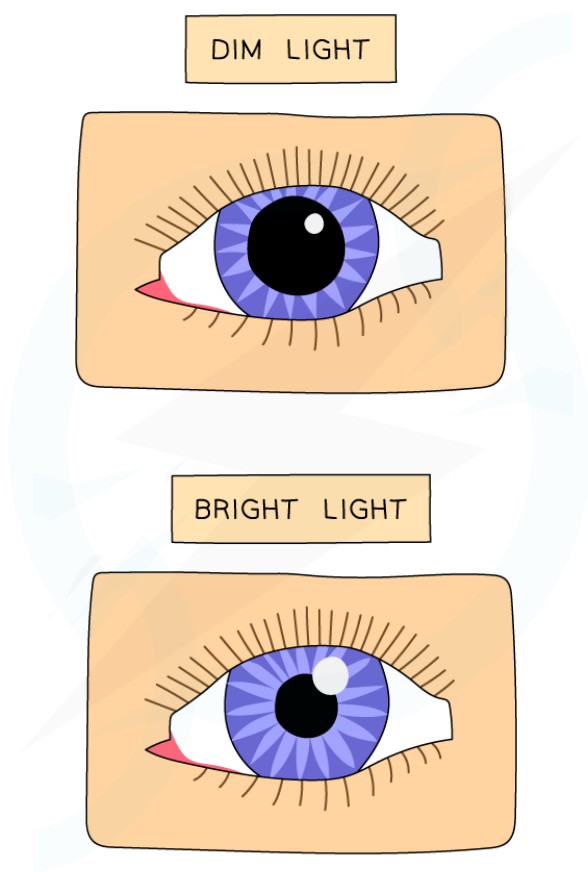
14.3 SENSE ORGANS: THE EYE

YOUR NOTES



The Pupil Reflex

- This is a **reflex action** carried out to **protect the retina from damage** in bright light and protect us from not seeing objects in dim light
- In **dim light** the pupil **dilates** (widens) in order to allow as much light into the eye as possible
- In **bright light** the pupil **constricts** (narrows) in order to prevent too much light entering the eye and damaging the retina



The pupil reflex

- In dim light, the pupil dilates (becomes larger) to allow more light to enter the eye to improve vision
- In bright light, the pupil constricts (gets smaller) to allow less light to enter the eye to protect the retina from damage



14 COORDINATION & RESPONSE

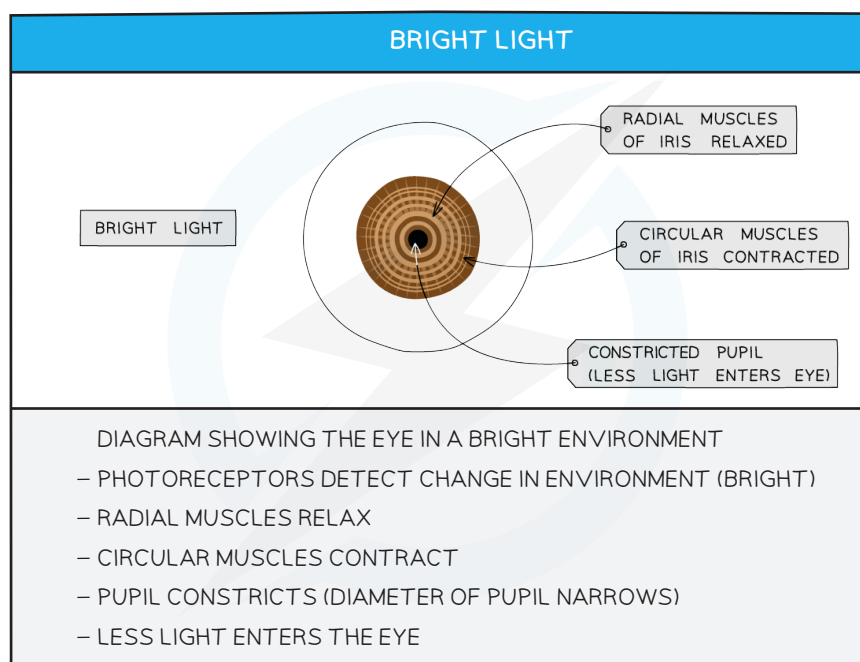
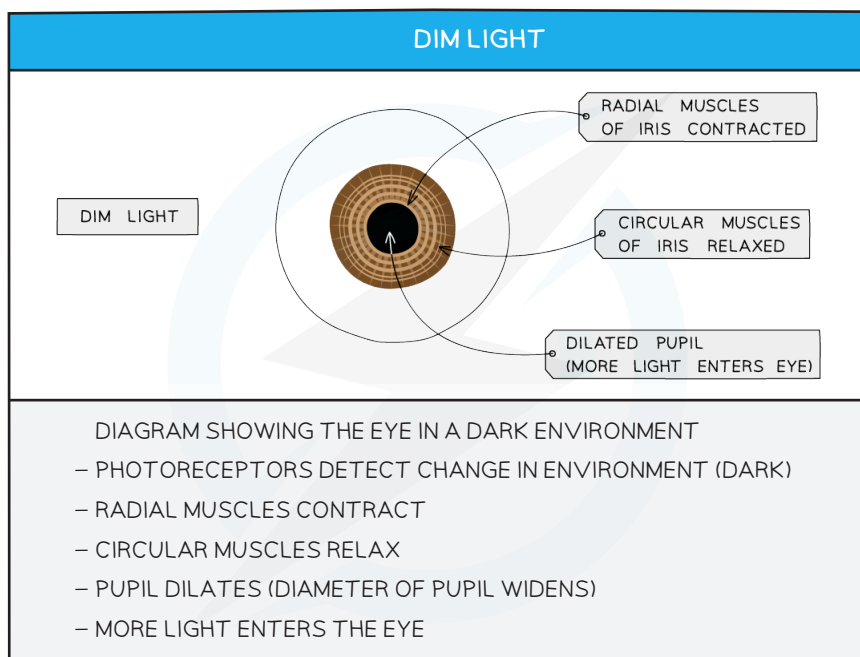
14.3 SENSE ORGANS: THE EYE cont...

YOUR NOTES



EXTENDED ONLY

How Does the Pupil Reflex Work?





14 COORDINATION & RESPONSE

14.3 SENSE ORGANS: THE EYE cont...

YOUR NOTES



EXTENDED ONLY cont...

STIMULUS	RADIAL MUSCLES	CIRCULAR MUSCLES	PUPIL SIZE	AMOUNT OF LIGHT ENTERS
DARK LIGHT	CONTRACTED	RELAXED	WIDE	MORE
BRIGHT LIGHT	RELAXED	CONTRACTED	NARROW	LESS

Accommodation: Viewing Near & Distant Objects

- The way the lens brings about fine focusing is called **accommodation**
- The lens is elastic and its shape can be changed when the **suspensory ligaments** attached to it become tight or loose
- The changes are brought about by the contraction or relaxation of the **ciliary muscles**

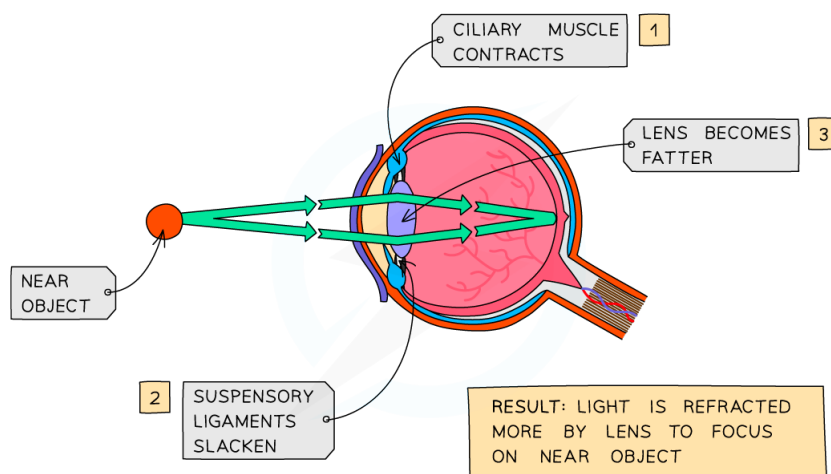


Diagram showing the eye when an object is close up

- Ciliary muscles contract
- Suspensory ligaments slacken
- This allows lens to become fatter
- Light is refracted more



14 COORDINATION & RESPONSE

14.3 SENSE ORGANS: THE EYE cont...

YOUR NOTES



EXTENDED ONLY cont...

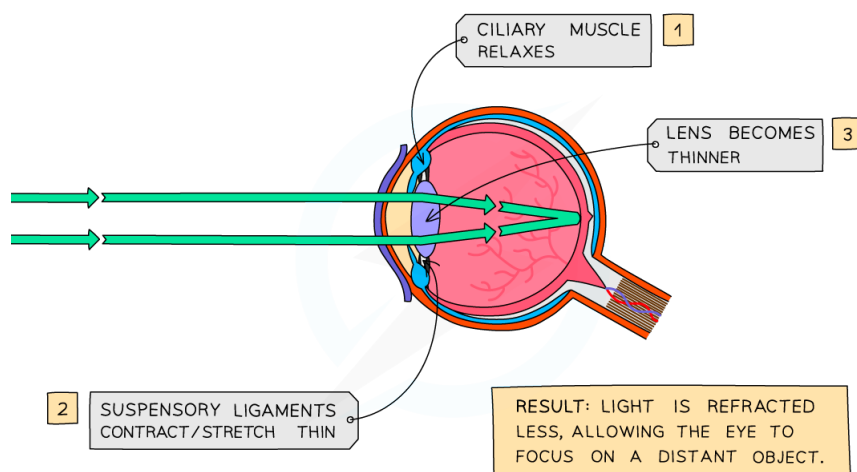


Diagram showing the eye when an object is far away

- Ciliary muscles relax
- Suspensory ligaments stretch thin
- This allows lens to become thinner
- Light is refracted less

	OBJECT FAR AWAY — THE LIGHT IS REFRACTED LESS	OBJECT CLOSE BY — — THE LIGHT IS REFRACTED MORE
CILIARY MUSCLES	RELAXED	CONTRACTED
SUSPENSORY LIGAMENTS	PULLED TIGHT	SLACK
LENS	THINNER	FATTER



14 COORDINATION & RESPONSE

14.3 SENSE ORGANS: THE EYE cont...



EXTENDED ONLY cont...

Rods & Cones

- There are two types of receptor cells in the retina:
- **Rods** which are sensitive to **dim light**
- **Cones** which distinguish between **different colours** in bright light
- There are **3 types of cone cells** which are sensitive to **different colours of light** (red, blue and green)
- The **fovea** is an area on the retina where **almost all of the cone cells are found**
- **Rod cells** are found **all over the retina**, other than the area where the **optic nerve attaches to the retina** – there are no light-sensitive cells at all in this area, and so it is known as the **blind spot**

YOUR NOTES



14.4 HORMONES IN HUMANS

What is a Hormone?

- A **hormone** is a chemical substance produced by a gland and carried by the blood, which alters the activity of one or more specific target organs i.e. they are chemicals which transmit information from one part of the organism to another and bring about a change
- The glands that produce hormones in animals are known collectively as the **endocrine system**
- Endocrine glands have a **good blood supply** as when they make hormones they need to get them into the bloodstream (specifically the blood plasma) as soon as possible so they can travel around the body to the target organs to bring about the response
- Once a hormone has been used, it is destroyed by the **liver**



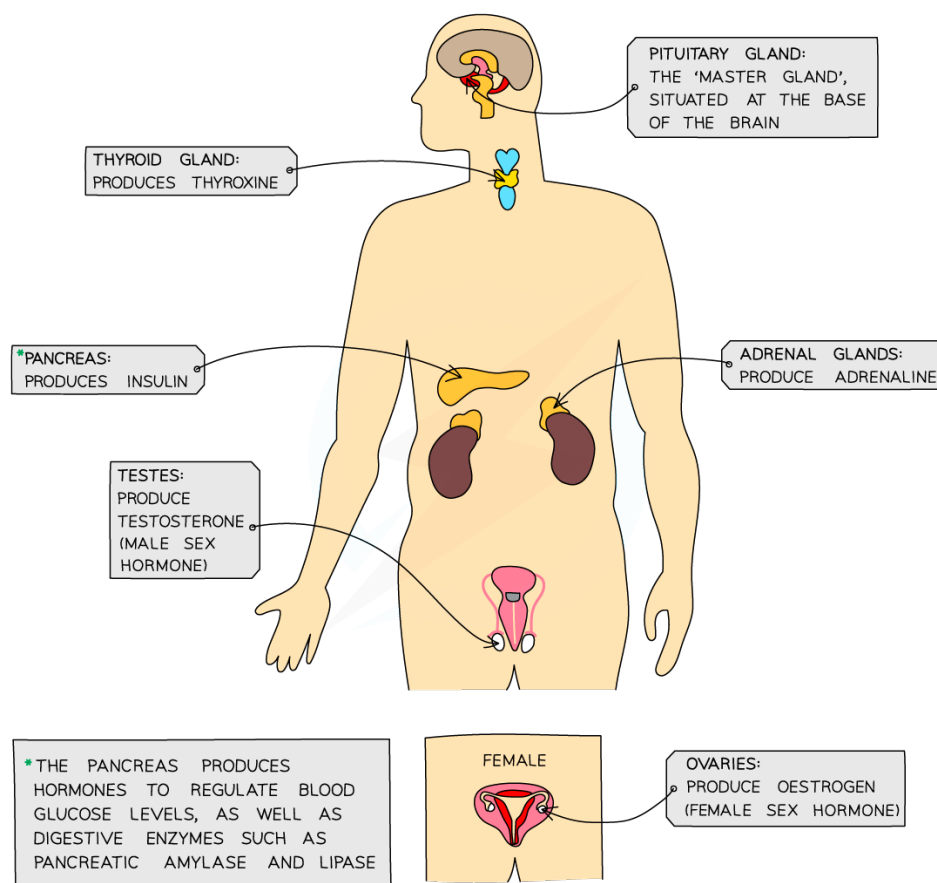
14 COORDINATION & RESPONSE

14.4 HORMONES IN HUMANS cont...

YOUR NOTES



The Endocrine System



The major endocrine glands in the body

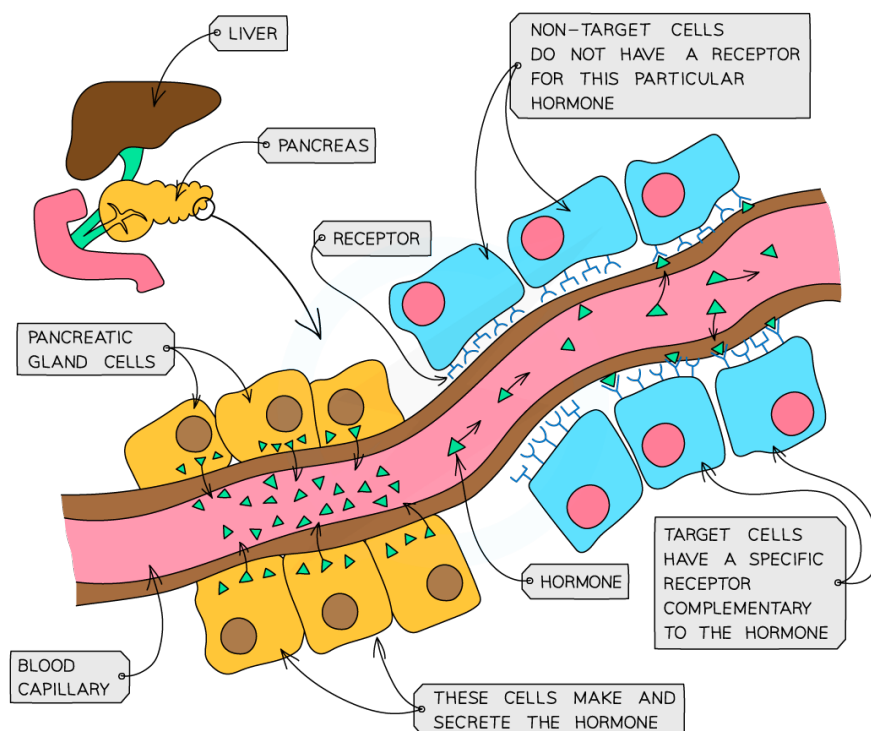
- Endocrine glands have a **good blood supply** as when they make hormones they need to get them into the bloodstream (specifically the blood plasma) as soon as possible so they can travel around the body to the target organs to bring about the response
- Hormones only affect cells with target receptors that the hormone can bind to. These are either found on the cell membrane, or inside cells. Receptors have to be complementary to hormones for their to be an effect
- The liver regulates levels of hormones in the blood; transforming or breaking down any that are in excess



14 COORDINATION & RESPONSE

14.4 HORMONES IN HUMANS cont...

YOUR NOTES



How hormones work

Important hormones in the human body

HORMONE	SOURCE	ROLE	EFFECT
ADRENALINE	ADRENAL GLAND	READIES THE BODY FOR A 'FIGHT OR FLIGHT' RESPONSE	INCREASES HEART AND BREATHING RATE, DILATES PUPILS
INSULIN	PANCREAS	LOWERS BLOOD GLUCOSE LEVELS	CAUSES EXCESS GLUCOSE IN THE BLOOD TO BE TAKEN UP BY THE MUSCLES AND LIVER AND CONVERTED INTO GLYCOGEN FOR STORAGE
TESTOSTERONE	TESTES	MAIN SEX HORMONE IN MALES	DEVELOPMENT OF SECONDARY SEXUAL CHARACTERISTICS IN MALES
OESTROGEN	OVARIES	MAIN SEX HORMONE IN FEMALES	DEVELOPMENT OF SECONDARY SEXUAL CHARACTERISTICS IN FEMALES AND CONTROLS MENSTRUAL CYCLE



14 COORDINATION & RESPONSE

14.4 HORMONES IN HUMANS cont...

YOUR NOTES



EXTENDED ONLY

More about Adrenaline

- Adrenaline is known as the fight or flight hormone as it is **produced in situations where the body may be in danger**
- It causes a range of different things to happen in the body, all designed to prepare it for movement (ie fight or flight).
- These include:
- **Increasing blood glucose concentration for increased respiration in muscle cells**
- **Increasing pulse rate and breathing rate** so glucose and oxygen can be **delivered to muscle cells**, and carbon dioxide taken away, from muscles cells **more quickly**
- **Diverting blood flow towards muscles** and away from non-essential parts of the body such as the alimentary canal; again to ensure the reactants of respiration are as available as possible
- **Dilating pupils** to allow as much light as possible to reach the retina so **more information can be sent to the brain**

Comparison of Nervous & Hormonal Control

	NERVOUS SYSTEM	ENDOCRINE SYSTEM
MADE UP OF:	NERVES (NEURONES), BRAIN, SPINAL CORD	GLANDS
TYPE OF MESSAGE:	ELECTRICAL IMPULSE	CHEMICAL HORMONE
SPEED OF TRANSMISSION:	VERY FAST	SLOWER
LENGTH OF EFFECT:	SHORT – UNTIL NERVE IMPULSES STOP	LONGER – UNTIL HORMONE IS BROKEN DOWN



14 COORDINATION & RESPONSE

14.4 HORMONES IN HUMANS cont...



EXAM TIP

Learning the list of effects of adrenaline on the body as it is a fairly common exam question and can be worth several easy marks.

YOUR NOTES



14.5 HOMEOSTASIS: BLOOD GLUCOSE

Homeostasis: Basics

- Homeostasis is defined as **the maintenance of a constant internal environment**



EXTENDED ONLY

Homeostasis

- "The maintenance of a constant internal environment" means that **internal conditions** within your body (such as temperature, blood pressure, water concentration, glucose concentration etc) **need to be kept within set limits** to ensure that reactions in body cells can function, and therefore the organism as a whole, can live
- When one of these conditions deviates far away from the normal if not brought back within set limits the **body will not function properly** and the eventual consequence without medical intervention will be death
- This is why diabetics need to control glucose intake (as their body cannot regulate it for them); why an extremely high and prolonged fever will kill you; or why drinking too little or too much water can damage cells throughout the body – especially the kidneys and brain – and lead to death within days
- Most homeostatic mechanisms in the body are controlled by a process known as **negative feedback**



14 COORDINATION & RESPONSE

14.5 HOMEOSTASIS: BLOOD GLUCOSE cont...

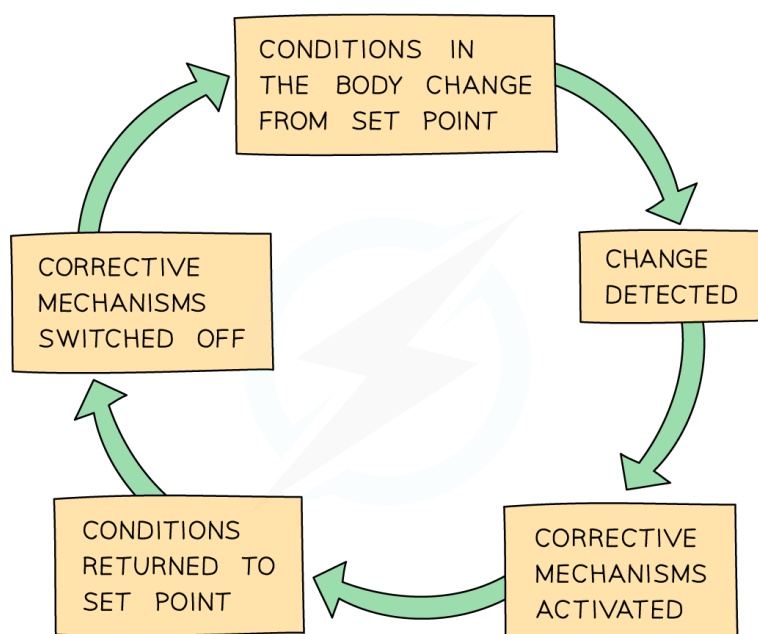
YOUR NOTES



EXTENDED ONLY cont...

Negative Feedback

- Negative feedback occurs when conditions change from the ideal or **set point**, and returns conditions to this set point
- It works in the following way:
- if the level of something **rises**, control systems are switched on to **reduce it** again
- if the level of something **falls**, control systems are switched on to **raise it** again
- Negative feedback mechanisms are usually a continuous cycle of bringing levels down and then bringing them back up so that overall, they stay within a **narrow range** of what is considered '**normal**'



The negative feedback cycle



14 COORDINATION & RESPONSE

14.5 HOMEOSTASIS: BLOOD GLUCOSE cont...

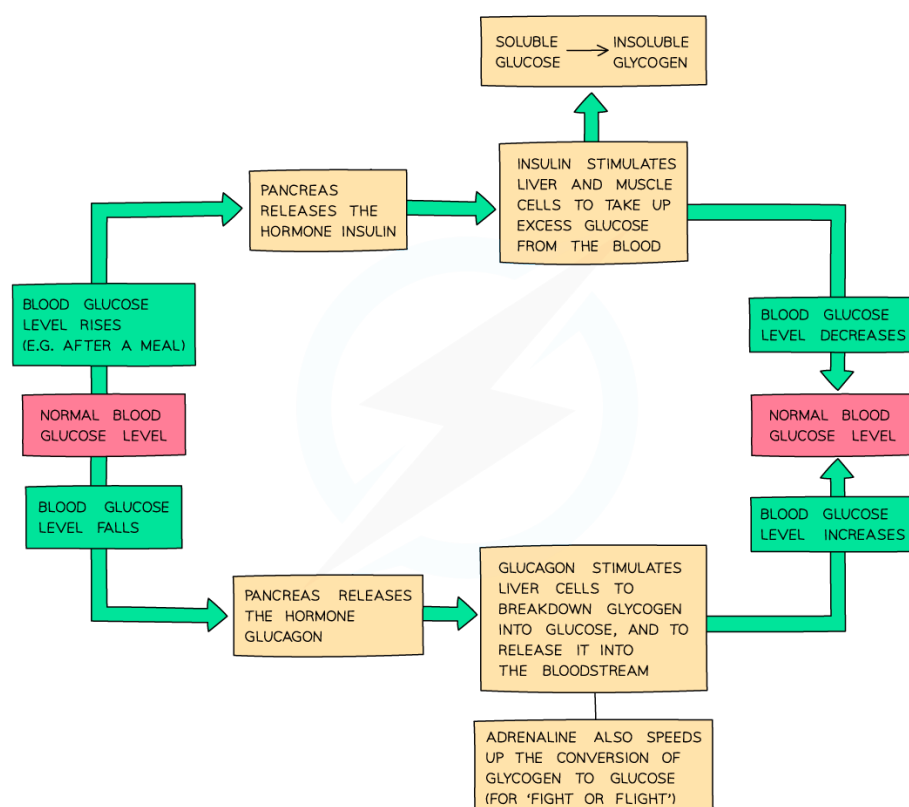
YOUR NOTES



EXTENDED ONLY cont...

Control of Blood Glucose Levels

- Blood glucose levels are controlled by a negative feedback mechanism involving the production of two **hormones** – **insulin** and **glucagon**
- Both hormones which control blood glucose concentration are made in the **pancreas**
- Insulin** is produced when **blood glucose rises** and **stimulates liver and muscle cells to convert excess glucose into glycogen to be stored**
- Glucagon** is produced when **blood glucose falls** and **stimulates liver and muscle cells to convert stored glycogen into glucose to be released into the blood**



Negative feedback regulation of blood glucose levels



14 COORDINATION & RESPONSE

14.5 HOMEOSTASIS: BLOOD GLUCOSE cont...

YOUR NOTES



EXTENDED ONLY cont...

Diabetes

- **Type 1 diabetes** is a condition where the blood glucose levels are not able to be regulated as the **insulin-secreting cells in the pancreas are not able to produce insulin**
- This means that blood glucose levels are often **far too high**
- It can be treated by **injecting insulin**
- The extra insulin causes the liver to convert **glucose into glycogen**, which **reduces** the blood glucose level
- **Symptoms** of diabetes include extreme thirst, weakness or tiredness, blurred vision, weight loss and loss of consciousness in extreme cases
- People with Type 1 diabetes have to **monitor** their blood glucose levels throughout the day as their levels of **physical activity** and their **diet** affect the amount of insulin needed
- They can help to **control** their blood glucose level by being careful with their **diet**
 - eating foods that will not cause large increases in blood glucose level, and by **exercising**, which can lower blood glucose levels due to increased respiration in the muscles



EXAM TIP

The terms glucagon and glycogen are very often mixed up by students as they sound similar. Remember:

- Glucagon is the **hormone**
- Glycogen is the polysaccharide that **glucose is stored as**

Learn the differences between the spellings and what each one does so you do not get confused in the exam!



14 COORDINATION & RESPONSE

14.6 HOMEOSTASIS: TEMPERATURE CONTROL

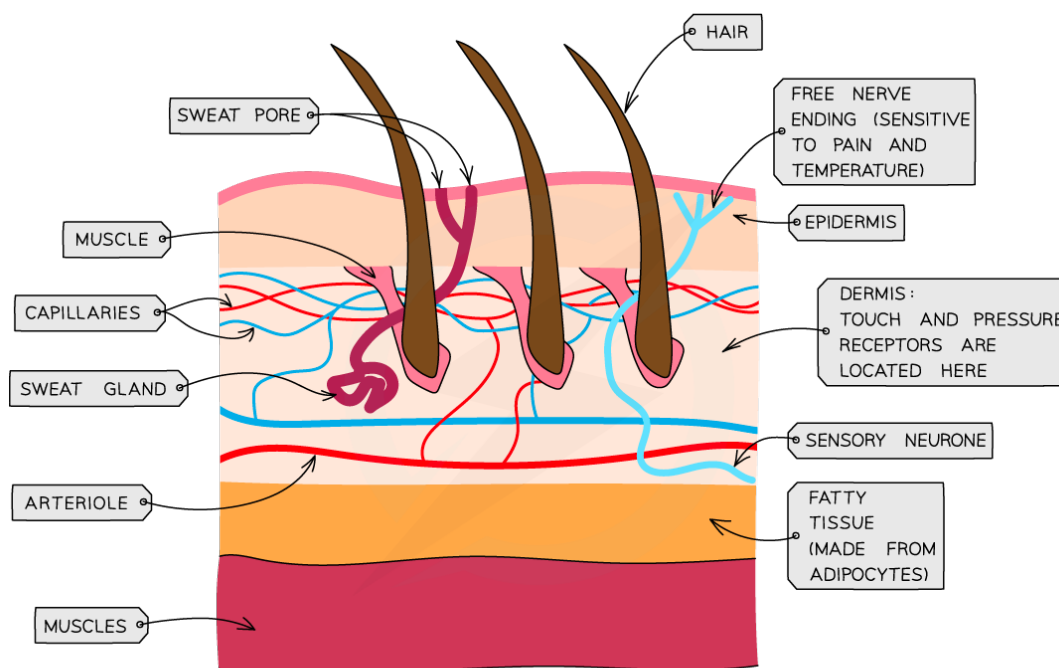
YOUR NOTES



The Skin & Homeostasis

- Control of body temperature is a **homeostatic** mechanism
- Homeostasis is **the maintenance of a constant internal environment**
- This means that **internal conditions within your body** (such as temperature, blood pressure, water concentration, glucose concentration etc) need to be **kept within set limits** in order to ensure that reactions in body cells can function and therefore the organism as a whole can live
- The human body maintains the temperature at which enzymes work best, around **37°C**
- If body temperature increases over this temperature, **enzymes will denature** and become less effective at catalysing reactions such as respiration

Structure of the Skin



A cross-section of human skin



14 COORDINATION & RESPONSE

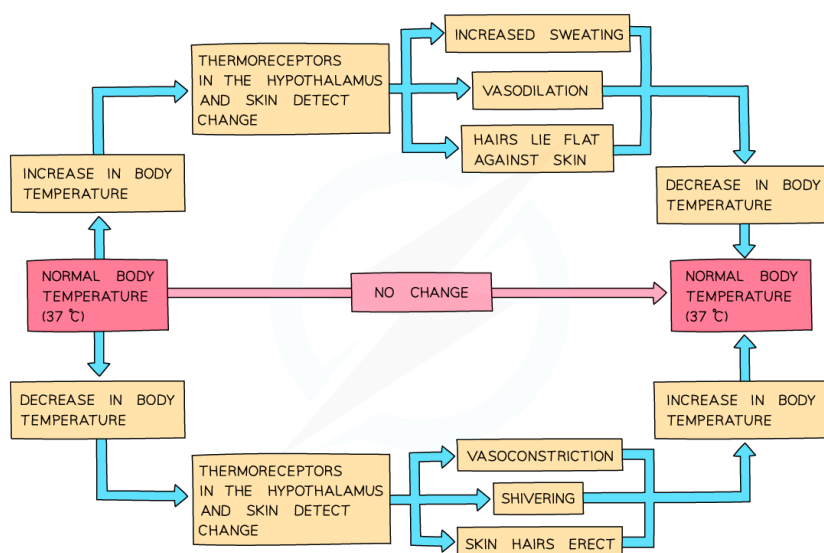
14.6 HOMEOSTASIS: TEMPERATURE CONTROL cont...

YOUR NOTES



Regulating Temperature: Basics

- Regulation is controlled by the **brain** which contains **receptors** sensitive to the temperature of the blood
- The **skin** also has **temperature receptors** and sends nervous impulses to the brain via **sensory neurones**
- The brain responds to this information by sending nerve impulses to **effectors** in the skin to **maintain the temperature within a narrow range of the optimum, 37°C**
- Fatty tissue** under the dermis acts as a layer of **insulation** to prevent too much body heat being lost through the skin



Homeostatic responses to changes in body temperature

WHEN WE ARE HOT	WHEN WE ARE COLD
SWEAT IS SECRETED BY SWEAT GLANDS THIS COOLS SKIN BY EVAPORATION. HEAT ENERGY FROM THE BODY IS LOST AS LIQUID WATER IN SWEAT BECOMES WATER VAPOUR (A STATE CHANGE).	SKELETAL MUSCLES CONTRACT RAPIDLY AND WE SHIVER. THESE INVOLUTARY MUSCLE CONTRACTIONS NEED ENERGY FROM RESPIRATION AND SOME OF THIS IS RELEASED AS HEAT.
HAIRS LIE FLAT AGAINST THE SKIN, ALLOWING AIR TO FREELY CIRCULATE. THIS INCREASES HEAT TRANSFER TO ENVIRONMENT BY RADIATION.	ERECT HAIRS TRAP A LAYER OF AIR AROUND THE SKIN WHICH ACTS AS AN INSULATOR, PREVENTING HEAT LOSS BY RADIATION.



14 COORDINATION & RESPONSE

14.6 HOMEOSTASIS: TEMPERATURE CONTROL cont...

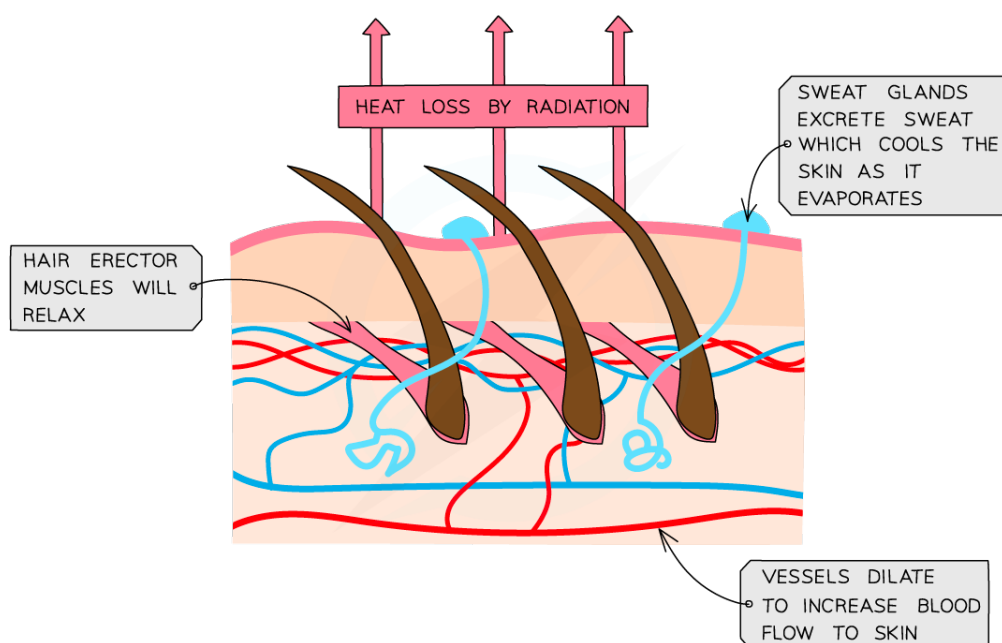
YOUR NOTES



EXTENDED ONLY

Vasodilation & Vasoconstriction

- When we are **cold** blood flow in capillaries slows down because arterioles leading to the skin capillaries get narrower – this is known as **vasoconstriction**
- This **reduces the amount of heat lost from blood by radiation** as less blood flows through the surface of the skin
- When we are **hot** blood flow in capillaries increases because blood vessels to the skin capillaries get wider – this is known as **vasodilation**
- This cools the body as blood (which carries heat around the body) is flowing at a faster rate through the skin's surface and so **more heat is lost by radiation**



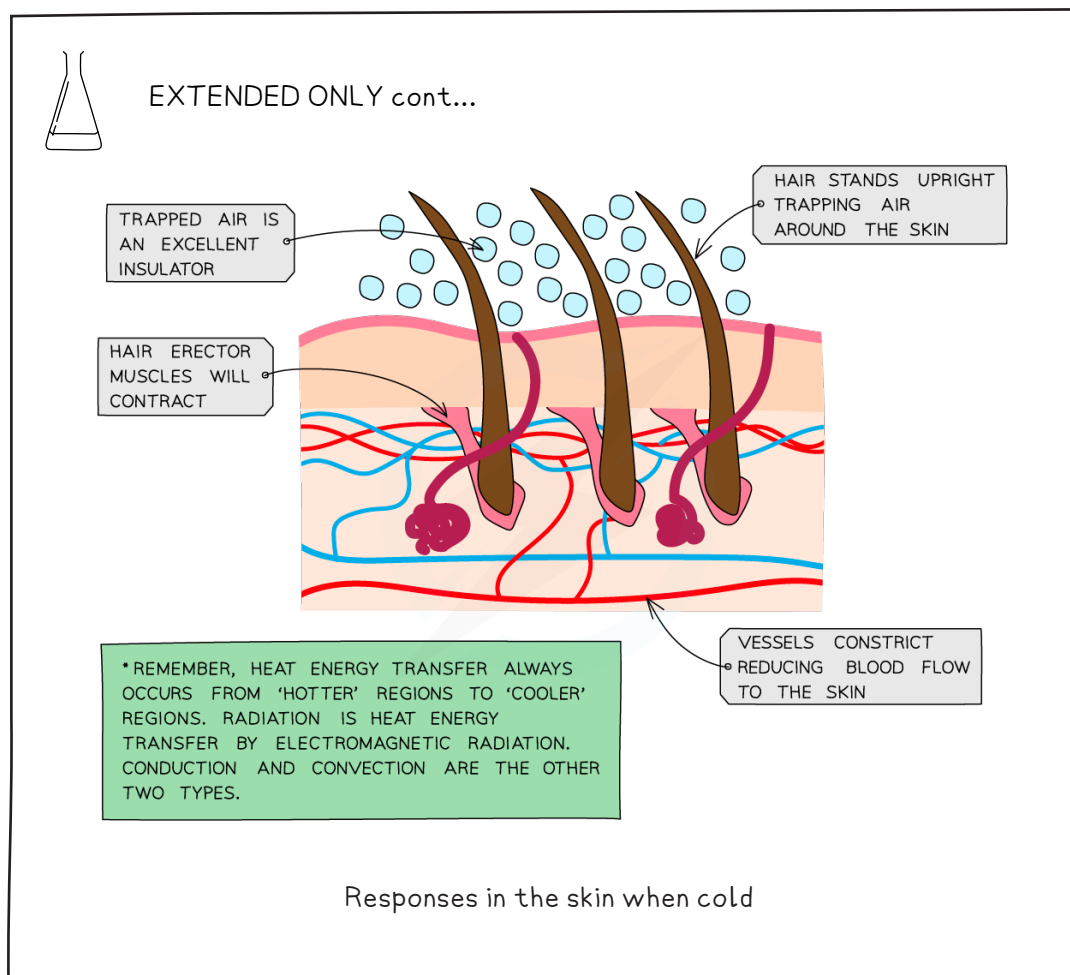
Responses in the skin when hot



14 COORDINATION & RESPONSE

14.6 HOMEOSTASIS: TEMPERATURE CONTROL cont...

YOUR NOTES



14.7 TROPIC RESPONSES

Plant Tropisms

- Plants can respond to changes in environment (stimuli) for survival, e.g. **light, water, gravity**
- Their responses are usually **much slower** than animals
- They grow either **towards a stimulus** (known as a positive response) or **away from a stimulus** (known as a negative response)
- The responses are known as **tropisms**



14 COORDINATION & RESPONSE

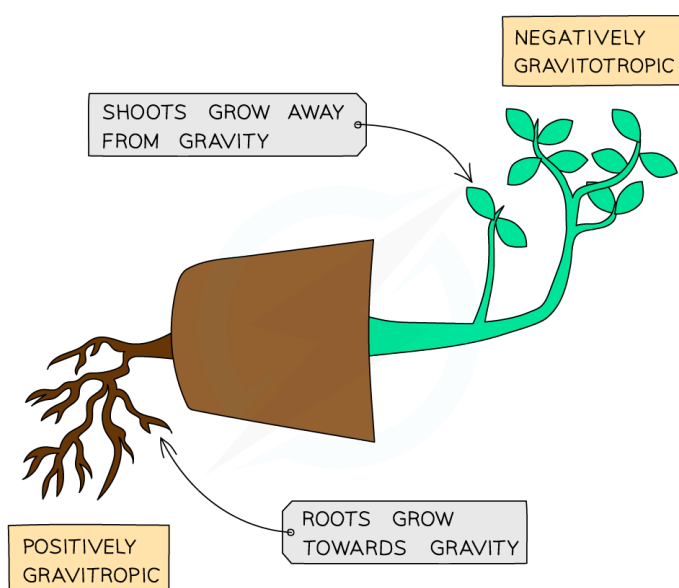
14.7 TROPIC RESPONSES cont...

YOUR NOTES



STIMULUS	NAME OF RESPONSE	DEFINITION	POSITIVE RESPONSE	NEGATIVE RESPONSE
GRAVITY	GRAVITROPISM (SOMETIMES CALLED GEOTROPISM)	GROWTH TOWARDS OR AWAY FROM GRAVITY	GROWTH TOWARDS GRAVITY (EG ROOTS)	GROWTH AWAY FROM GRAVITY (EG SHOOTS)
LIGHT	PHOTOTROPISM	GROWTH TOWARDS OR AWAY FROM THE DIRECTION OF LIGHT	GROWTH TOWARDS LIGHT (EG SHOOTS)	GROWTH AWAY FROM LIGHT (EG ROOTS)

- It is very important to a plant that its roots and shoots grow in the right directions
- Shoots must grow **upwards**, away from gravity and towards light, so that leaves are able to absorb sunlight
- This means that shoots have a **positive phototropic response** and a **negative gravitropic response**
- Roots need to grow **downwards** into the soil, away from light and towards gravity, in order to anchor the plant and absorb water and minerals from the soil particles.
- This means that roots have a **negative phototropic response** and a **positive gravitropic response**



Placing a plant on its side shows the gravitropic responses (also known as geotropic responses)



14 COORDINATION & RESPONSE

14.7 TROPIC RESPONSES cont...

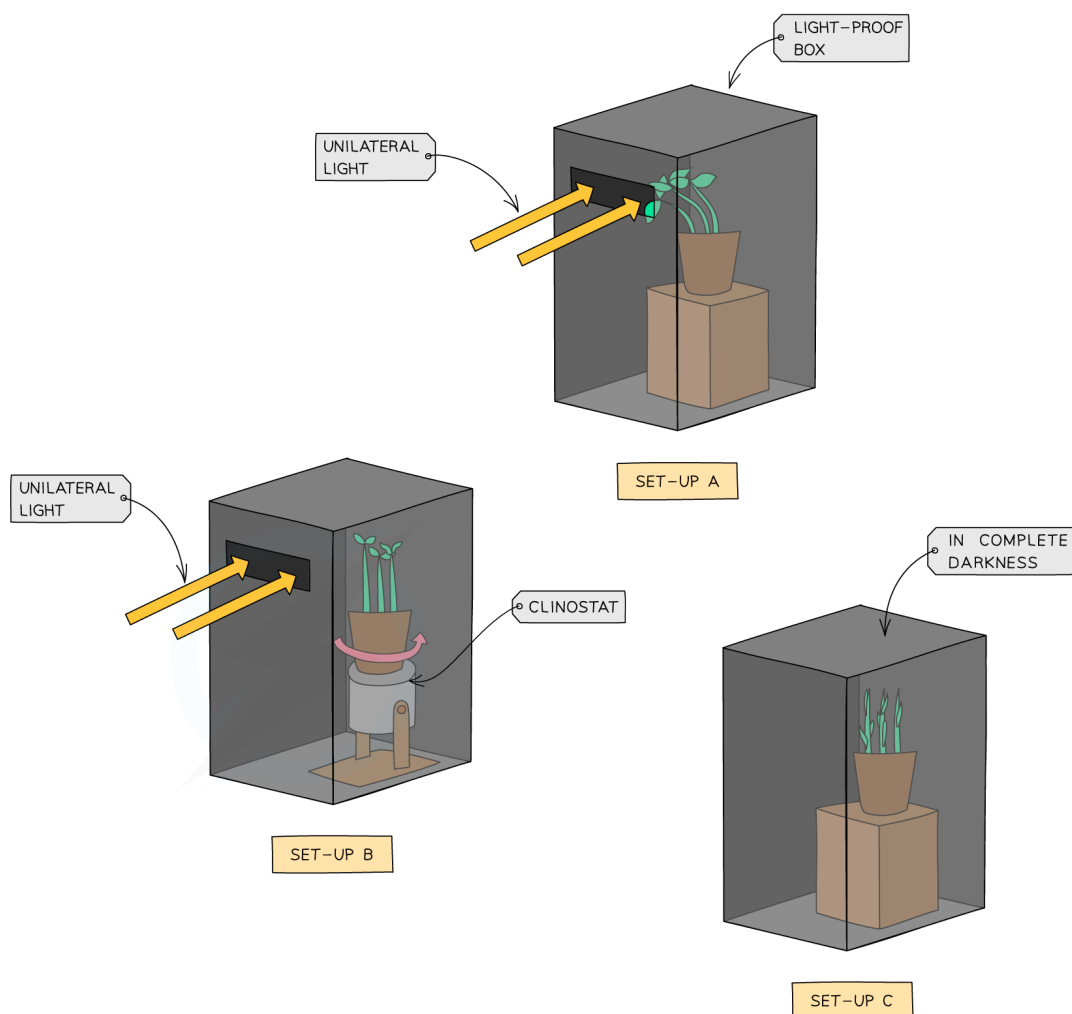
YOUR NOTES



Investigating Tropisms

Phototropisms

- Three identical plants are set up as shown below (A, B and C)



Investigating the phototropic response

- The seedlings in A grow **towards** the light source
- In B the effect of the light only coming from one direction has been **cancelled out** by using a **clinostat** (it revolves slowly and repeatedly so the shoots are evenly exposed to light)
- This means all sides of the seedlings get an **equal amount of light** so they do not curve towards the light source but grow straight up
- In C the seedlings grow straight up **looking for light** and the plant becomes tall and slender with yellowing leaves due to the lack of light



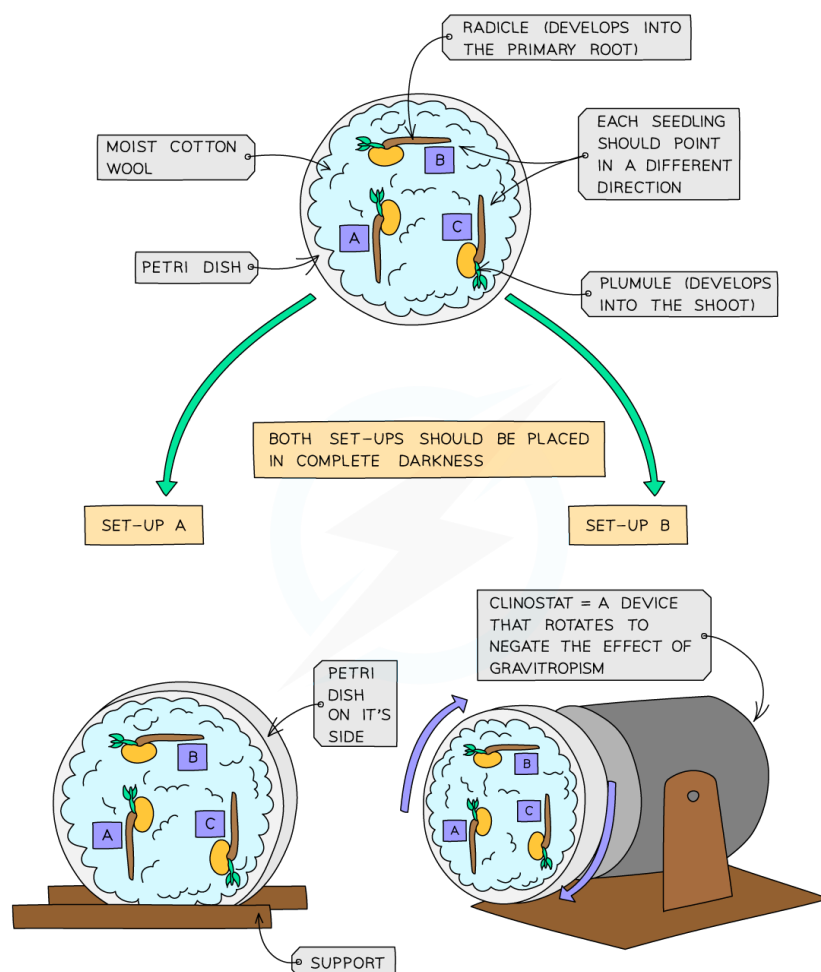
14 COORDINATION & RESPONSE

14.7 TROPIC RESPONSES cont...

YOUR NOTES



Gravitropism



Investigating the gravitropic response (set-up)

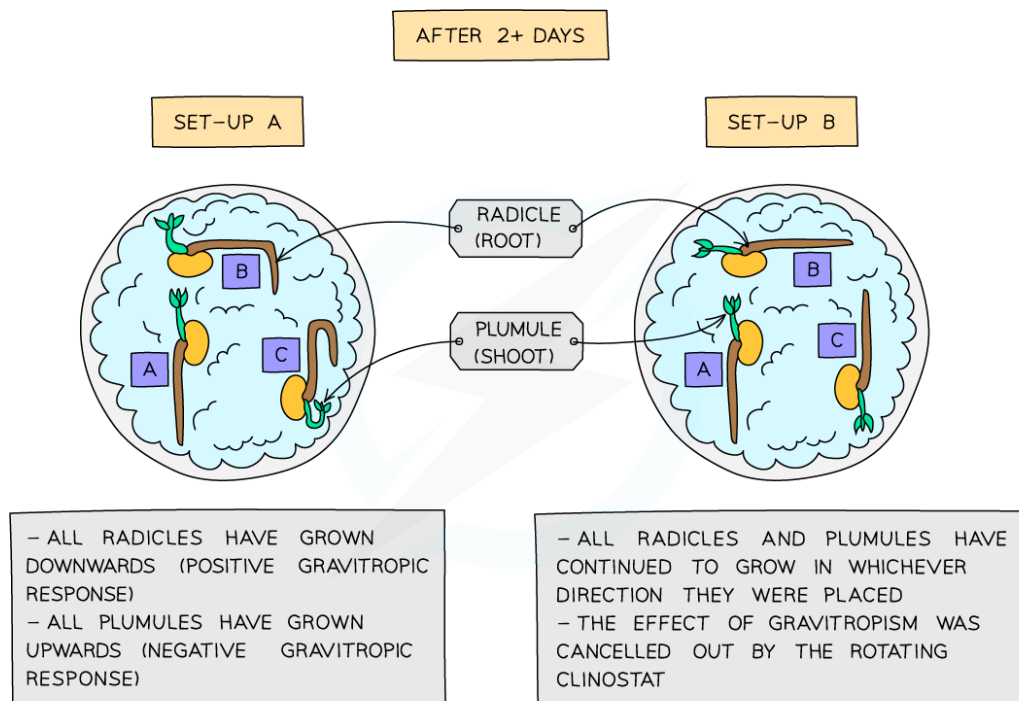
- Add some damp cotton wool to two petri dishes
- Place 3 bean seedlings in the cotton wool in each petri dish
 - A – radicle facing **downwards**
 - B – horizontally
 - C – radicle (root grows from here) facing **upwards**
- Cover each dish with a lid
- Attach one petri dish to a support so that it's on its side
- Attach the second petri dish to a clinostat (as shown in the diagrams above).
- Place both in a **light-proof box** (so that the seedlings are in complete darkness), leave for two days and then observe growth of the seedlings



14 COORDINATION & RESPONSE

14.7 TROPIC RESPONSES cont...

YOUR NOTES



Investigating the gravitropic response (results)

- In the first petri dish **all radicles (roots) have grown downwards** (positive gravitropic response) regardless of which way they were initially facing (horizontal, up or down) and **all plumules (shoots) have grown upwards** (negative gravitropic response)
- In the second petri dish, all radicles and all plumules have all grown **neither up nor down** but straight outwards **in whichever direction they were placed** as the **effect of gravity** has been **cancelled out by the revolving of the clinostat** – they have shown no gravitropic response at all
- The experiment needs to be done in a lightproof box in order to **cancel out the effect of light on the growth of the seedlings**



EXAM TIP

Know what a clinostat is and what it does (cancel out the effect of light or gravity).



14 COORDINATION & RESPONSE

14.7 TROPIC RESPONSES cont...

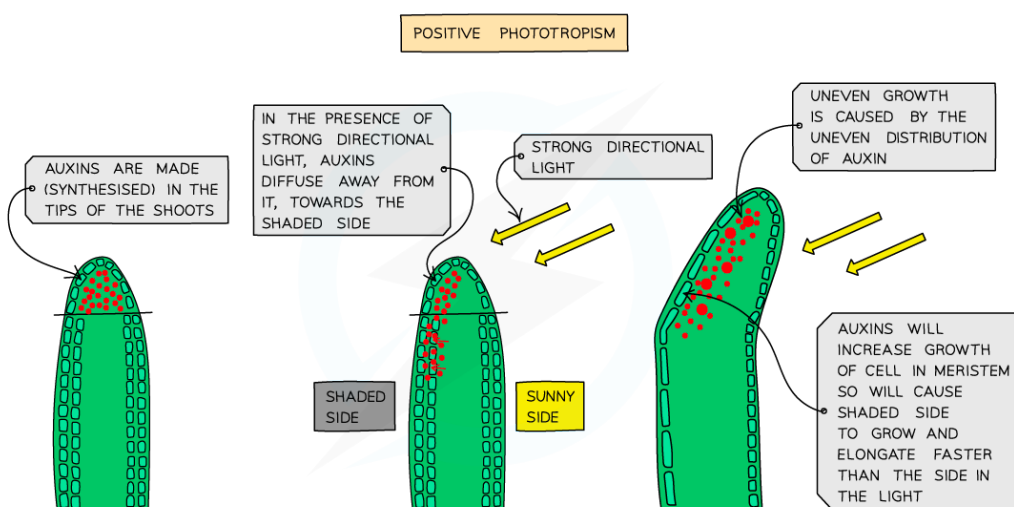
YOUR NOTES



EXTENDED ONLY

The Role of Auxin

- Plants respond to stimuli by producing a **growth hormone called auxin** which controls the direction of growth of roots or stems
- Therefore we say plants control their growth chemically
- Auxin is mostly made in the **tips** of the growing stems and roots and can diffuse to other parts of the stems or roots; spreading from a high concentration in the shoot tips down the shoot to an area of lower concentration
- Auxin stimulates the cells behind the tip **to elongate** (get larger); the more auxin there is, the faster they will elongate and grow
 - This is an important point. Only the region behind the tip of a shoot is able to contribute to growth by cell division and cell elongation. This part of a shoot is called the meristem
- If light shines all around the tip, auxin is distributed evenly throughout and the cells in the meristem grow at the same rate – this is what normally happens with plants growing outside
- When light shines on the shoot predominantly from one side though, the auxin produced in the tip **concentrates on the shaded side**, making the cells on that side elongate and grow faster than the cells on the sunny side
- This unequal growth on either side of the shoot **causes the shoot to bend** and grow in the direction of the light



Positive phototropism in plant shoots



14 COORDINATION & RESPONSE

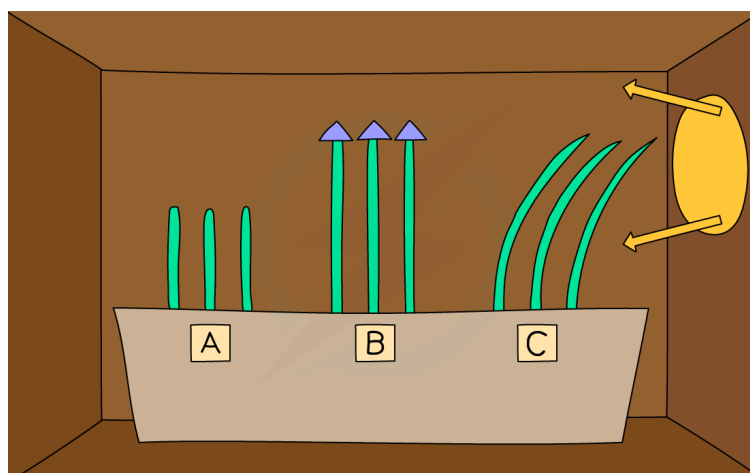
14.7 TROPIC RESPONSES cont...

YOUR NOTES



EXTENDED ONLY cont...

The role of auxin can be tested using seedlings placed in a box that has a slit on one side, only allowing light in from one direction:



Investigating the phototropic response set-up

	SEEDLING A	SEEDLING B	SEEDLING C
TREATMENT	THE TIPS OF THE STEMS HAVE BEEN REMOVED	NO LIGHT REACHES THE TIPS	MORE LIGHT REACHES ONE SIDE OF THE TIPS
EFFECT ON AUXIN CONCENTRATION	NO AUXIN IS PRODUCED	EQUAL CONCENTRATION OF AUXIN ON BOTH SIDES OF THE TIP	GREATER CONCENTRATION OF AUXIN ON SHADED SIDE
RESULT	THE STEMS DO NOT GROW LONGER	THE STEMS GROW EVENLY AND LONGER ON BOTH SIDES	THE CELLS ON THE DARKER SIDE OF THE STEMS GROW LONGER AND FASTER THAN CELLS IN THE LIGHT
REASON	THIS PROVES THAT AUXIN IS MADE IN THE TIP AND IS NEEDED TO STIMULATE CELL ELONGATION AND THEREFORE GROWTH. WITHOUT THE TIP, AND THEREFORE WITHOUT AUXIN, THERE IS NO FURTHER GROWTH IN THE STEM	THIS PROVES THAT IT IS THE EFFECT OF LIGHT ON AUXIN THAT CAUSES PHOTOTROPISM. THERE IS NO UNEVEN DISTRIBUTION OF AUXIN THROUGHOUT THE SHOOT TIP, SO GROWTH IS EVEN	THIS PROVES THAT SHOOTS GROW TOWARDS THE LIGHT BECAUSE THERE IS AN UNEVEN DISTRIBUTION OF AUXIN IN THE SHOOT TIP, CAUSED BY AUXIN'S RESPONSE TO LIGHT



14 COORDINATION & RESPONSE

14.7 TROPIC RESPONSES cont...

YOUR NOTES



EXTENDED ONLY cont...

Plant Hormones & Weedkillers

- Most **weedkillers** contain **synthetic hormones** like auxin – known as **2,4D**
- They are **selective** so they are sprayed onto an area such as a lawn or farm crops and the synthetic auxin **affects the weeds but not the grass / crop plants**
- The weeds respond by **growing very fast** and then **dying**, leaving more **space, nutrients and water** for the grass or crop plants to grow



EXAM TIP

You should be able to explain the results of an **experiment** into the effect of light on the growth of shoots.

Your syllabus does NOT require you to know anything about the effect of light (or gravity) on the growth of roots so don't spend time learning anything about it!

> NOW TRY SOME EXAM QUESTIONS



14 COORDINATION & RESPONSE

EXAM QUESTIONS

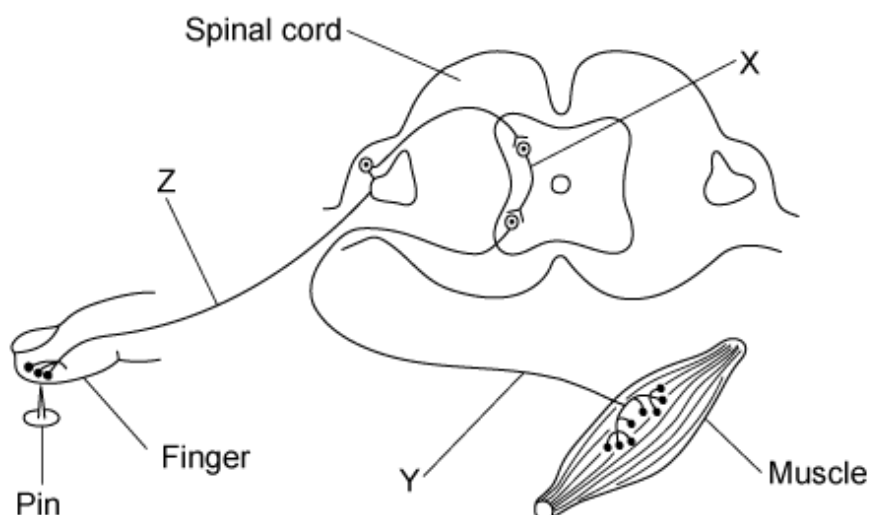
YOUR NOTES



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QUESTION 1

The image below shows a simple reflex arc.



Which of the following sequences correctly shows the order of cells that an impulse passes through during a reflex action?

	first	→	last
A	Z	X	Y
B	X	Y	Z
C	Y	X	Z
D	Z	Y	X



14 COORDINATION & RESPONSE

EXAM QUESTIONS cont...

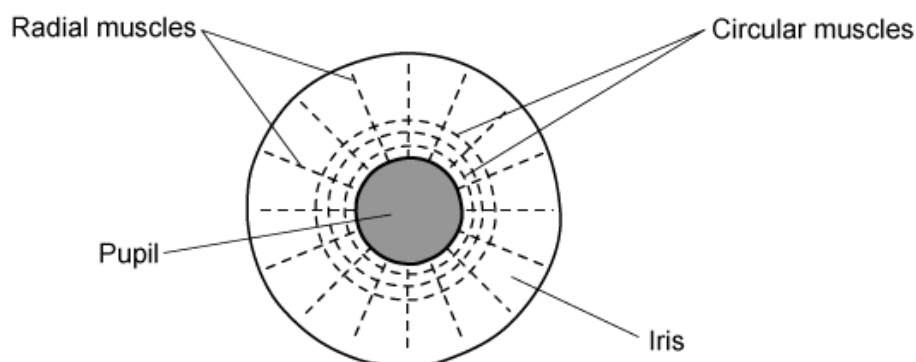
YOUR NOTES



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QUESTION 2

The image below shows some of the structures in the eye.



The circular and radial muscles can control the size of the pupil in the eye.

How would they respond in the eye of a person walking from a brightly lit room into a dark one?

	radial muscles	circular muscles	pupil
A	relaxes	contracts	constricts
B	relaxes	contracts	dilated
C	contracts	relaxes	dilated
D	contracts	relaxes	constricts



14 COORDINATION & RESPONSE

EXAM QUESTIONS cont...

YOUR NOTES



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QUESTION 3

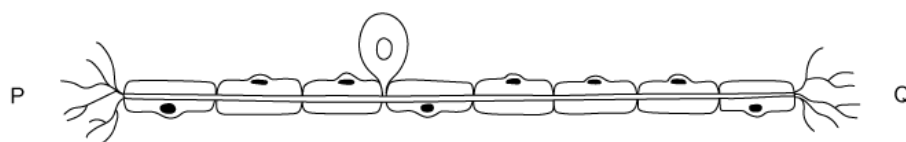
Which row of the following table correctly describes what happens in the body after a person consumes a can of sugary drink?

	role of the pancreas	role of the liver	effect
A	releases glucagon	converts glycogen into glucose	blood sugar levels rise
B	releases insulin	converts excess glucose into glycogen	blood sugar level falls
C	releases glucagon	converts excess glucose into glycogen	blood sugar level falls
D	releases insulin	converts glycogen into glucose	blood sugar levels rise

?

QUESTION 4

The image below shows a cell from the nervous system. The cell forms part of a reflex arc.



An impulse moves along the neurone from Q to P.

What structures would be found at P and Q?

	P	Q
A	receptor	relay neurone
B	relay neurone	receptor
C	motor neurone	relay neurone
D	gland	relay neurone



14 COORDINATION & RESPONSE

EXAM QUESTIONS cont...

YOUR NOTES



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QUESTION 5

An athlete takes part in a marathon.

What would the effect of an athlete taking part in a marathon race on a hot, sunny day?

	volume of water reabsorbed in the kidneys	volume of water lost from the skin	effect on hairs on the skin
A	less	decreases	lie flat
B	more	decreases	erect
C	less	increases	lie flat
D	more	increases	lie flat

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for more questions and revision notes